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Last login: Wed Jun 24 18:19:56 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % ls -l "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/result
s/"
total 5448
-rw-r--r--@ 1 stephenrajva  staff    22846 24 Jun 15:38 minimd_mesh_2x2x2_ppn1_nodes8_stats.csv
-rw-r--r-- 1 stephenrajva  staff   49604 24 Jun 15:39 minimd_mesh_2x2x4_ppn1_nodes16_stats.csv
-rw-r--r-- 1 stephenrajva  staff  227822 24 Jun 15:39 minimd_mesh_4x4x4_ppn1_nodes64_stats.csv
-rw-r--r-- 1 stephenrajva  staff  471518 24 Jun 15:40 minimd_mesh_4x4x8_ppn1_nodes128_stats.csv
-rw-r--r-- 1 stephenrajva  staff 2006259 24 Jun 15:40 minimd_mesh_8x8x8_ppn1_nodes512_stats.csv
stephenrajva@Stephens-MacBook-Air-4 ~ % cat "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/torus/torus.py"
import sst
from sst.merlin.base import *
from sst.merlin.endpoint import *
from sst.merlin.interface import *
from sst.merlin.topology import *
from sst.ember import *

PlatformDefinition.setCurrentPlatform("firefly-defaults")

# Topology: 3D Mesh (2x2x4) with 1 host per router (total 16 nodes)
topo = topoTorus()
topo.shape = "2x2x4"
topo.local_ports = 1
topo.width = "1x1x1"
topo.link_latency = "25ns"

# Router configuration
router = hr_router()
router.link_bw = "12GB/s"
router.flit_size = "16B"
router.xbar_bw = "24GB/s"
router.input_latency = "20ns"
router.output_latency = "20ns"
router.input_buf_size = "16kB"
router.output_buf_size = "16kB"
router.num_vns = 2
router.xbar_arb = "merlin.xbar_arb_lru"
topo.router = router

# Network interface configuration
nic = ReorderLinkControl()
nic.link_bw = "12GB/s"
nic.input_buf_size = "16kB"
nic.output_buf_size = "16kB"

# Create the system and assign the topology
system = System()
system.setTopology(topo)

# Job Patterns Based on MiniMD TAU Profiling
job = {
    "size": 16,
    "start": 0,
    "motifs": [
        {"pattern": "AllPingPong", "params": "iterations=325 messageSize=20510 computetime=476000"}, # Neighbor 1 (e.g., x+1)
        {"pattern": "AllPingPong", "params": "iterations=325 messageSize=9000 computetime=476000"}, # Neighbor 2 (e.g., y+1)
        {"pattern": "AllPingPong", "params": "iterations=325 messageSize=20510 computetime=476000"}, # Neighbor 3 (e.g., z+1)
        {"pattern": "AllPingPong", "params": "iterations=325 messageSize=9000 computetime=476000"}, # Neighbor 4 (e.g., diagonal)
    ]
    {"pattern": "Allreduce", "params": "count=1 iterations=40 compute=2451000"},
    {"pattern": "Barrier", "params": "iterations=15 compute=1392000"},
}

# Assign jobs to system
ep = EmberMPIJob(job["start"], job["size"])
ep.network_interface = nic
ep.addMotif("Init")
for motif in job["motifs"]:
    ep.addMotif(f"{motif['pattern']} {motif['params']}")
ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"
system.allocateNodes(ep, "linear")

# Build the system
system.build()

# Configure statistics and debugging
sst.setStatisticLoadLevel(12)
sst.setStatisticOutput("sst.statOutputCSV", {
    "filepath": "minimd_torus_stats.csv",
    "separator": ",",
})

# Enable key statistics
sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {
    "MotifLog": "1"
})
sst.enableAllStatisticsForComponentType("ember.nic")

# Print confirmation
print("MiniMD Pattern Simulation Configured:")
print("- Topology: 3D torus 2x2x4")

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print("- Hosts per router: 1")
print(f"- Total compute nodes: {2*2*4*topo.local_ports}")
print(f"- Total ranks: {job['size']}")
print(f"- Jobs configured: 1 job, {job['size']} total ranks")%
henrajva@Stephens-MacBook-Air-4 ~ %
stephenrajva@Stephens-MacBook-Air-4 ~ % ls -l "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D/result
s/"
total 5448
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-rw-r--r-- 1 stephenrajva  staff    227822 24 Jun 15:39 minimd_mesh_4x4x4_ppn1_nodes64_stats.csv
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from sst.ember import *

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topo.link_latency = "25ns"

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router.output_latency = "20ns"
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router.output_buf_size = "16kB"
router.num_vns = 2
router.xbar_arb = "merlin.xbar_arb_lru"
topo.router = router

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ep.addMotif("Fini")
ep.nic.nic2host_lat = "100ns"
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system.build()

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sst.setStatisticLoadLevel(12)
sst.setStatisticOutput("sst.statOutputCSV", {
    "filepath": "minimd_torus_stats.csv",
    "separator": ",",
})

# Enable key statistics
sst.enableAllStatisticsForComponentType("merlin.hr_router")
sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
sst.enableAllStatisticsForComponentType("ember.EmberEngine", {
    "MotifLog": "1"
})
sst.enableAllStatisticsForComponentType("ember.nic")

```

```

# Print confirmation
print("MiniMD Pattern Simulation Configured:")
print("- Topology: 3D torus 2x2x4")
print("- Hosts per router: 1")
print(f"- Total compute nodes: {2*2*4*topo.local_ports}")
print(f"- Total ranks: {job['size']}")
print(f"- Jobs configured: 1 job, {job['size']} total ranks")
henrajva@Stephens-MacBook-Air-4 ~ % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/mesh3D"
export DATA_DIR="./results"
python3 analyze_minimd_mesh3d.py

```

Topology	Avg latency (ns)	Avg hop count
2x2x2	143.09	1.714
2x2x4	615.45	2.400
4x4x4	615.94	3.810
4x4x8	681.99	5.165
8x8x8	683.13	7.890

```

Saved: minimd_latency_vs_size.png
Saved: minimd_hop_count_vs_size.png
stephenrajva@Stephens-MacBook-Air-4 mesh3D % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/torus"
chmod +x run_minimd_torus_sweep.sh

# test the smallest, check PPN=1
export MESH_SHAPE="2x2x2"
export STATS_DIR="./results"
sst minimd_torus3d_ppn1.py
chmod: run_minimd_torus_sweep.sh: No such file or directory
zsh: command not found: #
Failed to canonicalize path [minimd_torus3d_ppn1.py]: No such file or directory
stephenrajva@Stephens-MacBook-Air-4 torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/torus"
chmod +x run_minimd_torus_sweep.sh

# test the smallest, check PPN=1
export MESH_SHAPE="2x2x2"
export STATS_DIR="./results"
sst minimd_torus3d_ppn1.py
chmod: run_minimd_torus_sweep.sh: No such file or directory
zsh: command not found: #
Failed to canonicalize path [minimd_torus3d_ppn1.py]: No such file or directory
stephenrajva@Stephens-MacBook-Air-4 torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/miniapps/MiniMD_simulation/torus"
ls -l minimd_torus3d_ppn1.py run_minimd_torus_sweep.sh
ls: minimd_torus3d_ppn1.py: No such file or directory
ls: run_minimd_torus_sweep.sh: No such file or directory
stephenrajva@Stephens-MacBook-Air-4 torus %
stephenrajva@Stephens-MacBook-Air-4 torus %
stephenrajva@Stephens-MacBook-Air-4 torus % # if loose in Downloads:
mv ~/Downloads/minimd_torus3d_ppn1.py ~/Downloads/run_minimd_torus_sweep.sh "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/"

# if inside a files-N folder instead:
mv ~/Downloads/files-*/minimd_torus3d_ppn1.py ~/Downloads/files-*/run_minimd_torus_sweep.sh "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/"
zsh: command not found: #
mv: /Users/stephenrajva/Downloads/minimd_torus3d_ppn1.py: No such file or directory
mv: /Users/stephenrajva/Downloads/run_minimd_torus_sweep.sh: No such file or directory
zsh: command not found: #
zsh: no matches found: /Users/stephenrajva/Downloads/files-*/minimd_torus3d_ppn1.py
stephenrajva@Stephens-MacBook-Air-4 torus % find ~/Downloads -name "minimd_torus3d_ppn1.py" 2>/dev/null
find ~/Downloads -name "run_minimd_torus_sweep.sh" 2>/dev/null
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/minimd_torus3d_ppn1.py
/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/run_minimd_torus_sweep.sh
stephenrajva@Stephens-MacBook-Air-4 torus % ./run_minimd_torus_sweep.sh
zsh: no such file or directory: ./run_minimd_torus_sweep.sh
stephenrajva@Stephens-MacBook-Air-4 torus % cp "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/ppn1/analyze_universal.py" .

python3 - << 'EOF'
import re
f = "analyze_universal.py"
s = open(f).read()
s = re.sub(r'STUDY\s*=\s*.*?$', 'STUDY = "MiniMD_Torus3D"', s, count=1)
s = re.sub(r'TOPOLOGY\s*=\s*.*?$', 'TOPOLOGY = "torus"', s, count=1)
new_runs = '''RUNS = [
    ("2x2x2", (2,2,2), "minimd_torus_2x2x2_ppn1_nodes8_stats.csv"),
    ("2x2x4", (2,2,4), "minimd_torus_2x2x4_ppn1_nodes16_stats.csv"),
    ("4x4x4", (4,4,4), "minimd_torus_4x4x4_ppn1_nodes64_stats.csv"),
    ("4x4x8", (4,4,8), "minimd_torus_4x4x8_ppn1_nodes128_stats.csv"),
    ("8x8x8", (8,8,8), "minimd_torus_8x8x8_ppn1_nodes512_stats.csv"),
]'''
s = re.sub(r'RUNS = \[.*?\]', new_runs, s, count=1, flags=re.DOTALL)
open(f, "w").write(s)
print("Updated for MiniMD 3D torus.")
EOF

export DATA_DIR="./results"
python3 analyze_universal.py
Updated for MiniMD 3D torus.
!! missing: ./results/minimd_torus_2x2x2_ppn1_nodes8_stats.csv (skipping)
!! missing: ./results/minimd_torus_2x2x4_ppn1_nodes16_stats.csv (skipping)
!! missing: ./results/minimd_torus_4x4x4_ppn1_nodes64_stats.csv (skipping)
!! missing: ./results/minimd_torus_4x4x8_ppn1_nodes128_stats.csv (skipping)
!! missing: ./results/minimd_torus_8x8x8_ppn1_nodes512_stats.csv (skipping)

```

```

MiniMD_Torus3D
Size      Latency(ns)  Hops    Thru(GB/s)  Packets  Bits          Idle%  Stalls
-----
Saved: MiniMD_Torus3D_latency_ns.png
Saved: MiniMD_Torus3D_hops.png
Saved: MiniMD_Torus3D_throughput.png
Saved: MiniMD_Torus3D_packets.png
Saved: MiniMD_Torus3D_bits.png
Saved: MiniMD_Torus3D_idle_pct.png
Saved: MiniMD_Torus3D_stalls.png
stephenrajva@Stephens-MacBook-Air-4 torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST-Stimulator/MiniMD/Torus"
ls -l minimd_torus3d_ppn1.py
-rw-r--r--@ 1 stephenrajva  staff  3679 24 Jun 20:20 minimd_torus3d_ppn1.py
stephenrajva@Stephens-MacBook-Air-4 Torus % export MESH_SHAPE="2x2x2"
export STATS_DIR="./results"
sst minimd_torus3d_ppn1.py
=====
Topology      : 3D Torus 2x2x2
Routers       : 8
Total nodes   : 8
Total MPI ranks : 8
PPN (ranks/node) : 1  <-- should be 1
Stats CSV     : ./results/minimd_torus_2x2x2_ppn1_nodes8_stats.csv
=====
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
161473537282:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
161473537282:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
161473537282:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
161473537282:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 161476.347 us, loop 325, bufLen 20510, latency 248.425 us. bandwidth 0.082560 GB/s
161476527798:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
161476527798:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
161476527798:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
161476527798:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
317171011766:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
317171011766:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
317171011766:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
317171011766:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 155695.826 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.037573 GB/s
317172353865:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
317172353865:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
317172353865:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
317172353865:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
478645711449:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=8
478645711449:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=8
478645711449:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=8
478645711449:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=8
AllPingPong: total time 161476.348 us, loop 325, bufLen 20510, latency 248.425 us. bandwidth 0.082560 GB/s
478648701965:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=8
478648701965:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=8
478648701965:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=8
478648701965:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=8
AllPingPong: total time 155695.827 us, loop 325, bufLen 9000, latency 239.532 us. bandwidth 0.037573 GB/s
Allreduce: ranks 8, loop 40, 1 double(s), latency 2456.323 us
Barrier: ranks 8, loop 15, latency 1397.398 us
Simulation is complete, simulated time: 753.566 ms
stephenrajva@Stephens-MacBook-Air-4 Torus % export STATS_DIR="./results"
export MESH_SHAPE="2x2x4"; sst minimd_torus3d_ppn1.py
export MESH_SHAPE="4x4x4"; sst minimd_torus3d_ppn1.py
export MESH_SHAPE="4x4x8"; sst minimd_torus3d_ppn1.py
export MESH_SHAPE="8x8x8"; sst minimd_torus3d_ppn1.py
=====
Topology      : 3D Torus 2x2x4
Routers       : 16
Total nodes   : 16
Total MPI ranks : 16
PPN (ranks/node) : 1  <-- should be 1
Stats CSV     : ./results/minimd_torus_2x2x4_ppn1_nodes16_stats.csv
=====
180000:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
180000:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
180000:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
180000:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
180000:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
180000:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
180000:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
180000:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
180000:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
180000:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
180000:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
180000:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
180000:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
180000:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
180000:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
180000:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162380668621:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
162380668621:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16

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162380668621:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
162380668621:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
162381118179:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
162381118179:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
162381118179:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
162381118179:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
162383592383:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
162383592383:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
162383592383:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
162383592383:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 162383.861 us, loop 325, bufLen 20510, latency 249.821 us. bandwidth 0.082099 GB/s
162384041557:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
162384041557:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
162384041557:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
162384041557:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
318232182598:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
318232182598:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
318232182598:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
318232182598:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
318232699806:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
318232699806:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
318232699806:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
318232699806:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
318233762149:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
318233762149:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
318233762149:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
318233762149:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 155850.238 us, loop 325, bufLen 9000, latency 239.770 us. bandwidth 0.037536 GB/s
318234279357:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
318234279357:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
318234279357:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
318234279357:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
480613738971:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=16
480613738971:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=16
480613738971:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=16
480613738971:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=16
480614188529:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=16
480614188529:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=16
480614188529:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=16
480614188529:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=16
480616662733:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=16
480616662733:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=16
480616662733:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=16
480616662733:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=16
AllPingPong: total time 162382.832 us, loop 325, bufLen 20510, latency 249.820 us. bandwidth 0.082099 GB/s
480617111907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=16
480617111907:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=16
480617111907:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=16
480617111907:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=16
AllPingPong: total time 155850.238 us, loop 325, bufLen 9000, latency 239.770 us. bandwidth 0.037536 GB/s
Allreduce: ranks 16, loop 40, 1 double(s), latency 2458.205 us
Barrier: ranks 16, loop 15, latency 1399.311 us
Simulation is complete, simulated time: 755.796 ms
=====
Topology      : 3D Torus 4x4x4
Routers       : 64
Total nodes   : 64
Total MPI ranks : 64
PPN (ranks/node) : 1  <-- should be 1
Stats CSV     : ./results/minimd_torus_4x4x4_ppn1_nodes64_stats.csv
=====
180000:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=64
180000:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=64
180000:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=64
180000:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=64
180000:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=64
180000:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=64
180000:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=64
180000:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=64
180000:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=64
180000:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=64
180000:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=64
180000:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=64
180000:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=64
180000:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=64
180000:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=64
180000:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=64
180000:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
180000:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
180000:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=64
180000:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=64
180000:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=64
180000:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=64
180000:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=64
180000:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=64
180000:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=64
180000:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=64
180000:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=64
180000:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=64
180000:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=64
180000:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=64
180000:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=64
180000:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=64
180000:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
180000:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
```

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```
480614188529:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=64
480614188529:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=64
480616662733:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=64
480616662733:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=64
480616662733:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=64
480616662733:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=64
480616662733:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=64
480616662733:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=64
480616662733:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=64
480616662733:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=64
480616662733:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=64
480616662733:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=64
480616662733:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=64
480616662733:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=64
480616662733:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=64
480616662733:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=64
480616662733:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=64
480616662733:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=64
AllPingPong: total time 162382.832 us, loop 325, bufLen 20510, latency 249.820 us. bandwidth 0.082099 GB/s
480617111907:0:0:EmberEngine:AllPingPong:generate:59: rank=0 size=64
480617111907:0:1:EmberEngine:AllPingPong:generate:59: rank=1 size=64
480617111907:0:2:EmberEngine:AllPingPong:generate:59: rank=2 size=64
480617111907:0:3:EmberEngine:AllPingPong:generate:59: rank=3 size=64
480617111907:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=64
480617111907:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=64
480617111907:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=64
480617111907:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=64
480617111907:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=64
480617111907:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=64
480617111907:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=64
480617111907:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=64
480617111907:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=64
480617111907:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=64
480617111907:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=64
480617111907:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=64
AllPingPong: total time 155850.238 us, loop 325, bufLen 9000, latency 239.770 us. bandwidth 0.037536 GB/s
Allreduce: ranks 64, loop 40, 1 double(s), latency 2462.436 us
Barrier: ranks 64, loop 15, latency 1403.584 us
Simulation is complete, simulated time: 756.035 ms
=====
Topology      : 3D Torus 4x4x8
Routers       : 128
Total nodes   : 128
Total MPI ranks : 128
PPN (ranks/node) : 1 <-- should be 1
Stats CSV     : ./results/minimd_torus_4x4x8_ppn1_nodes128_stats.csv
=====
180000:0:127:EmberEngine:AllPingPong:generate:59: rank=127 size=128
180000:0:126:EmberEngine:AllPingPong:generate:59: rank=126 size=128
180000:0:125:EmberEngine:AllPingPong:generate:59: rank=125 size=128
180000:0:124:EmberEngine:AllPingPong:generate:59: rank=124 size=128
180000:0:123:EmberEngine:AllPingPong:generate:59: rank=123 size=128
180000:0:122:EmberEngine:AllPingPong:generate:59: rank=122 size=128
180000:0:121:EmberEngine:AllPingPong:generate:59: rank=121 size=128
180000:0:120:EmberEngine:AllPingPong:generate:59: rank=120 size=128
180000:0:119:EmberEngine:AllPingPong:generate:59: rank=119 size=128
180000:0:118:EmberEngine:AllPingPong:generate:59: rank=118 size=128
180000:0:117:EmberEngine:AllPingPong:generate:59: rank=117 size=128
180000:0:116:EmberEngine:AllPingPong:generate:59: rank=116 size=128
180000:0:115:EmberEngine:AllPingPong:generate:59: rank=115 size=128
180000:0:114:EmberEngine:AllPingPong:generate:59: rank=114 size=128
180000:0:113:EmberEngine:AllPingPong:generate:59: rank=113 size=128
180000:0:112:EmberEngine:AllPingPong:generate:59: rank=112 size=128
180000:0:111:EmberEngine:AllPingPong:generate:59: rank=111 size=128
180000:0:110:EmberEngine:AllPingPong:generate:59: rank=110 size=128
180000:0:109:EmberEngine:AllPingPong:generate:59: rank=109 size=128
180000:0:108:EmberEngine:AllPingPong:generate:59: rank=108 size=128
180000:0:107:EmberEngine:AllPingPong:generate:59: rank=107 size=128
180000:0:106:EmberEngine:AllPingPong:generate:59: rank=106 size=128
180000:0:105:EmberEngine:AllPingPong:generate:59: rank=105 size=128
180000:0:104:EmberEngine:AllPingPong:generate:59: rank=104 size=128
180000:0:103:EmberEngine:AllPingPong:generate:59: rank=103 size=128
180000:0:102:EmberEngine:AllPingPong:generate:59: rank=102 size=128
180000:0:101:EmberEngine:AllPingPong:generate:59: rank=101 size=128
180000:0:100:EmberEngine:AllPingPong:generate:59: rank=100 size=128
180000:0:99:EmberEngine:AllPingPong:generate:59: rank=99 size=128
180000:0:98:EmberEngine:AllPingPong:generate:59: rank=98 size=128
180000:0:97:EmberEngine:AllPingPong:generate:59: rank=97 size=128
180000:0:96:EmberEngine:AllPingPong:generate:59: rank=96 size=128
180000:0:95:EmberEngine:AllPingPong:generate:59: rank=95 size=128
180000:0:94:EmberEngine:AllPingPong:generate:59: rank=94 size=128
180000:0:93:EmberEngine:AllPingPong:generate:59: rank=93 size=128
180000:0:92:EmberEngine:AllPingPong:generate:59: rank=92 size=128
180000:0:91:EmberEngine:AllPingPong:generate:59: rank=91 size=128
180000:0:90:EmberEngine:AllPingPong:generate:59: rank=90 size=128
180000:0:89:EmberEngine:AllPingPong:generate:59: rank=89 size=128
180000:0:88:EmberEngine:AllPingPong:generate:59: rank=88 size=128
180000:0:87:EmberEngine:AllPingPong:generate:59: rank=87 size=128
180000:0:86:EmberEngine:AllPingPong:generate:59: rank=86 size=128
180000:0:85:EmberEngine:AllPingPong:generate:59: rank=85 size=128
180000:0:84:EmberEngine:AllPingPong:generate:59: rank=84 size=128
180000:0:83:EmberEngine:AllPingPong:generate:59: rank=83 size=128
180000:0:82:EmberEngine:AllPingPong:generate:59: rank=82 size=128
180000:0:81:EmberEngine:AllPingPong:generate:59: rank=81 size=128
180000:0:80:EmberEngine:AllPingPong:generate:59: rank=80 size=128
```


[illegible]

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```
Total MPI ranks      : 512
PPN (ranks/node)    : 1  <-- should be 1
Stats CSV            : ./results/minimd_torus_8x8x8_ppn1_nodes512_stats.csv
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

480680632089:0:4:EmberEngine:AllPingPong:generate:59: rank=4 size=512
480680632089:0:5:EmberEngine:AllPingPong:generate:59: rank=5 size=512
480680632089:0:6:EmberEngine:AllPingPong:generate:59: rank=6 size=512
480680632089:0:7:EmberEngine:AllPingPong:generate:59: rank=7 size=512
480680632089:0:8:EmberEngine:AllPingPong:generate:59: rank=8 size=512
480680632089:0:9:EmberEngine:AllPingPong:generate:59: rank=9 size=512
480680632089:0:10:EmberEngine:AllPingPong:generate:59: rank=10 size=512
480680632089:0:11:EmberEngine:AllPingPong:generate:59: rank=11 size=512
480680632089:0:12:EmberEngine:AllPingPong:generate:59: rank=12 size=512
480680632089:0:13:EmberEngine:AllPingPong:generate:59: rank=13 size=512
480680632089:0:14:EmberEngine:AllPingPong:generate:59: rank=14 size=512
480680632089:0:15:EmberEngine:AllPingPong:generate:59: rank=15 size=512
480680632089:0:16:EmberEngine:AllPingPong:generate:59: rank=16 size=512
480680632089:0:17:EmberEngine:AllPingPong:generate:59: rank=17 size=512
480680632089:0:18:EmberEngine:AllPingPong:generate:59: rank=18 size=512
480680632089:0:19:EmberEngine:AllPingPong:generate:59: rank=19 size=512
480680632089:0:20:EmberEngine:AllPingPong:generate:59: rank=20 size=512
480680632089:0:21:EmberEngine:AllPingPong:generate:59: rank=21 size=512
480680632089:0:22:EmberEngine:AllPingPong:generate:59: rank=22 size=512
480680632089:0:23:EmberEngine:AllPingPong:generate:59: rank=23 size=512
480680632089:0:24:EmberEngine:AllPingPong:generate:59: rank=24 size=512
480680632089:0:25:EmberEngine:AllPingPong:generate:59: rank=25 size=512
480680632089:0:26:EmberEngine:AllPingPong:generate:59: rank=26 size=512
480680632089:0:27:EmberEngine:AllPingPong:generate:59: rank=27 size=512
480680632089:0:28:EmberEngine:AllPingPong:generate:59: rank=28 size=512
480680632089:0:29:EmberEngine:AllPingPong:generate:59: rank=29 size=512
480680632089:0:30:EmberEngine:AllPingPong:generate:59: rank=30 size=512
480680632089:0:31:EmberEngine:AllPingPong:generate:59: rank=31 size=512
480680632089:0:32:EmberEngine:AllPingPong:generate:59: rank=32 size=512
480680632089:0:33:EmberEngine:AllPingPong:generate:59: rank=33 size=512
480680632089:0:34:EmberEngine:AllPingPong:generate:59: rank=34 size=512
480680632089:0:35:EmberEngine:AllPingPong:generate:59: rank=35 size=512
480680632089:0:36:EmberEngine:AllPingPong:generate:59: rank=36 size=512
480680632089:0:37:EmberEngine:AllPingPong:generate:59: rank=37 size=512
480680632089:0:38:EmberEngine:AllPingPong:generate:59: rank=38 size=512
480680632089:0:39:EmberEngine:AllPingPong:generate:59: rank=39 size=512
480680632089:0:40:EmberEngine:AllPingPong:generate:59: rank=40 size=512
480680632089:0:41:EmberEngine:AllPingPong:generate:59: rank=41 size=512
480680632089:0:42:EmberEngine:AllPingPong:generate:59: rank=42 size=512
480680632089:0:43:EmberEngine:AllPingPong:generate:59: rank=43 size=512
480680632089:0:44:EmberEngine:AllPingPong:generate:59: rank=44 size=512
480680632089:0:45:EmberEngine:AllPingPong:generate:59: rank=45 size=512
480680632089:0:46:EmberEngine:AllPingPong:generate:59: rank=46 size=512
480680632089:0:47:EmberEngine:AllPingPong:generate:59: rank=47 size=512
480680632089:0:48:EmberEngine:AllPingPong:generate:59: rank=48 size=512
480680632089:0:49:EmberEngine:AllPingPong:generate:59: rank=49 size=512
480680632089:0:50:EmberEngine:AllPingPong:generate:59: rank=50 size=512
480680632089:0:51:EmberEngine:AllPingPong:generate:59: rank=51 size=512
480680632089:0:52:EmberEngine:AllPingPong:generate:59: rank=52 size=512
480680632089:0:53:EmberEngine:AllPingPong:generate:59: rank=53 size=512
480680632089:0:54:EmberEngine:AllPingPong:generate:59: rank=54 size=512
480680632089:0:55:EmberEngine:AllPingPong:generate:59: rank=55 size=512
480680632089:0:56:EmberEngine:AllPingPong:generate:59: rank=56 size=512
480680632089:0:57:EmberEngine:AllPingPong:generate:59: rank=57 size=512
480680632089:0:58:EmberEngine:AllPingPong:generate:59: rank=58 size=512
480680632089:0:59:EmberEngine:AllPingPong:generate:59: rank=59 size=512
480680632089:0:60:EmberEngine:AllPingPong:generate:59: rank=60 size=512
480680632089:0:61:EmberEngine:AllPingPong:generate:59: rank=61 size=512
480680632089:0:62:EmberEngine:AllPingPong:generate:59: rank=62 size=512
480680632089:0:63:EmberEngine:AllPingPong:generate:59: rank=63 size=512
AllPingPong: total time 155825.891 us, loop 325, bufLen 9000, latency 239.732 us. bandwidth 0.037542 GB/s
Allreduce: ranks 512, loop 40, 1 double(s), latency 2470.947 us
Barrier: ranks 512, loop 15, latency 1412.236 us
Simulation is complete, simulated time: 756.558 ms
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -l results/
total 6032
-rw-r--r-- 1 stephenrajva staff 33522 24 Jun 20:29 minimd_torus_2x2x2_ppn1_nodes8_stats.csv
-rw-r--r-- 1 stephenrajva staff 67178 24 Jun 20:29 minimd_torus_2x2x4_ppn1_nodes16_stats.csv
-rw-r--r-- 1 stephenrajva staff 269698 24 Jun 20:29 minimd_torus_4x4x4_ppn1_nodes64_stats.csv
-rw-r--r-- 1 stephenrajva staff 540380 24 Jun 20:29 minimd_torus_4x4x8_ppn1_nodes128_stats.csv
-rw-r--r-- 1 stephenrajva staff 2167991 24 Jun 20:30 minimd_torus_8x8x8_ppn1_nodes512_stats.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % export DATA_DIR="./results"
python3 analyze_universal.py
/opt/homebrew/Cellar/python@3.14/3.14.5/Frameworks/Python.framework/Versions/3.14/Resources/Python.app/Contents/MacOS/Python: can't
open file '/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus/analyze_universal.py': [Errno 2] No such file
or directory
stephenrajva@Stephens-MacBook-Air-4 Torus % cd "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniMD/Torus"

cp "/Users/stephenrajva/Downloads/HPC-Upload 2/SST_Stimulator/MiniFE/Mesh/ppn1/analyze_universal.py" .

python3 - << 'EOF'
import re
f = "analyze_universal.py"
s = open(f).read()
s = re.sub(r'STUDY\s*=\s*.*?$', 'STUDY = "MiniMD_Torus3D"', s, count=1)
s = re.sub(r'TOPOLOGY\s*=\s*.*?$', 'TOPOLOGY = "torus"', s, count=1)
new_runs = '''RUNS = [
    ("2x2x2", (2,2,2), "minimd_torus_2x2x2_ppn1_nodes8_stats.csv"),
    ("2x2x4", (2,2,4), "minimd_torus_2x2x4_ppn1_nodes16_stats.csv"),
    ("4x4x4", (4,4,4), "minimd_torus_4x4x4_ppn1_nodes64_stats.csv"),
    ("4x4x8", (4,4,8), "minimd_torus_4x4x8_ppn1_nodes128_stats.csv"),
    ("8x8x8", (8,8,8), "minimd_torus_8x8x8_ppn1_nodes512_stats.csv"),
]'''
s = re.sub(r'RUNS = \[.*?\]', new_runs, s, count=1, flags=re.DOTALL)
open(f, "w").write(s)

```